Received: 29 July 2019 | Revised: 31 May 2020 | Accepted: 5 June 2020

DOI: 10.1111/1541-4337.12602

COMPREHENSIVE REVIEWS IN FOOD SCIENCE AND FOOD SAFETY

Macrofungi: A review of cultivation strategies, bioactivity, and application of mushrooms

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Funding information
Natural Science Foundation of Zhejiang Province, Grant/Award Number: LR13C200002

Abstract

The production of macrofungi (mushrooms) as well as their economic value have been steadily increasing globally. The use of functional foods, dietary supplements, and traditional medicines derived from macrofungi is increasing as they have numerous health benefits as well as abundant nutrients. Macrofungi are diverse with complex and highly varied growth conditions and bioactive constituents, most macrofungal resources have not yet been fully explored and applicated, leading to an urgent need for appropriate strategies to address the problem. Increasing attention has been paid to the macrofungal cultivation and application, in particular, potential prebiotics. Herein, the present review comprehensively summarizes recent progress in the cultivation, newly identified bioactive constituents, and their effects on gut microbiota as well as the potential ways in which they affect human health. Moreover, the macrofungal food development is discussed to improve food nutritional value and change the quality characteristics of food. Finally, the review addresses consumer safety concerns and the prospective genetic manipulation of macrofungi. We hope that this review can provide a comprehensive research reference for ensuring the safety and efficacy, along with maximizing the value and profitability of macrofungi production.

KEYWORDS

bioactive compound, cultivation technology, gut microbiota, mushroom, nutritional and therapeutic applications

1 | INTRODUCTION

Macrofungi, including members of phylum Basidiomycota and phylum Ascomycota in the kingdom of Fungi, have morphologically diverse epigeous or hypogeous fruiting bodies (Govorushko, Rezaee, Dumanov, & Tsatsakis, 2019) (Figure 1) and are collectively referred to as mushrooms. Macrofungi degrade lignocellulosic and pectic compounds in the environment (Sumic et al., 2017). Certain macrofungi can serve as ectomycorrhizal (EM) fungus that forms symbiotic associations with plants (Lutzoni et al., 2018). Nearly 14,000 macrofungal species have been identified worldwide. To date, >2,000 edible and/or medicinal macrofungi have been characterized (Meenu & Xu, 2019; Zhang, Venkitasamy, Pan, & Wang, 2013). Macrofungi are nutritious and beneficial for human health. They have been called the "meat of the forest" or the "meat for the poor" in certain countries (Dimitrijevic et al., 2018). In fresh macrofungal fruiting bodies, the moisture content is approximately 90%. Based on dry matter, they are abundant carbohydrates (50% to 65%) and proteins (19% to 35%) with trace amounts of fats (2% to 6%) (Rathore, Prasad, & Sharma, 2017; Wang et al., 2014). Moreover, They also include several different types of bioactive

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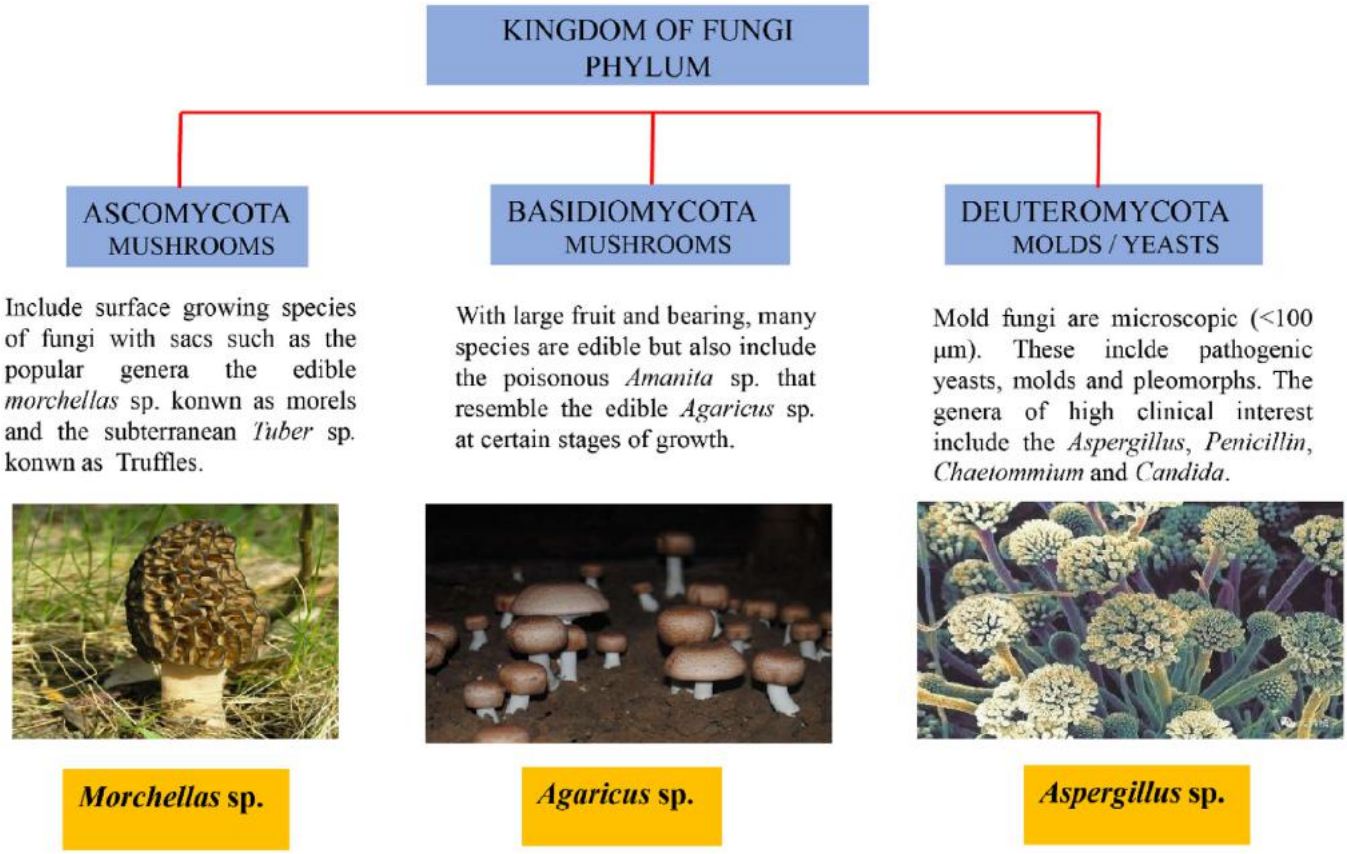
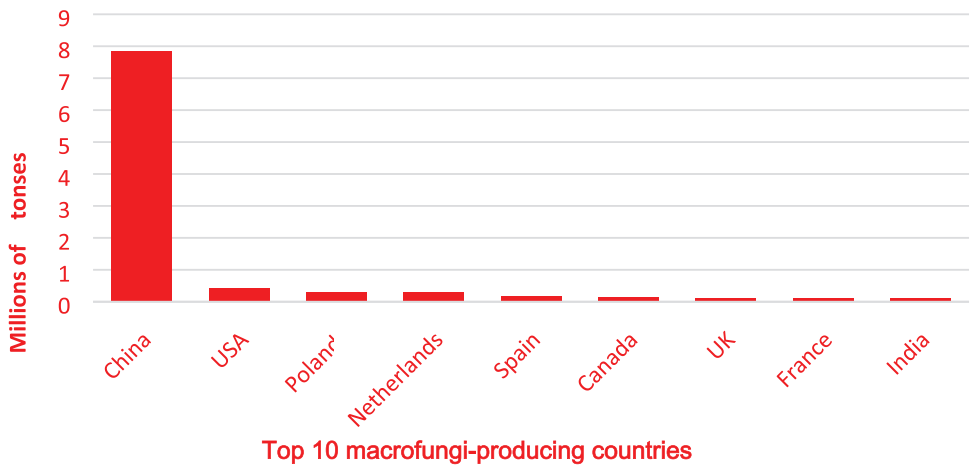


FIGURE 1 Kingdom Fungi. (Adapted with permission from Govorushko et al., 2019)

TABLE 1 Top ten macrofungi-producing countries (FAO, 2016)



constituents such as polysaccharides, dietary fiber, unsaturated fatty acids (palmitic, oleic, and linoleic), biologically active proteins (enzymes, lectins, and ergothioneine), phenolic compounds (phenolic acids and polyphenols), and vitamins (thiamine, riboflavin, ascorbic acid, niacin, and tocopherols). Hence, macrofungi are potentially important sources of low-calorie functional foods and nutraceuticals.

For millennia, macrofungi have been widely used in many regions (Chang & Wasser, 2012). China accounts for >80% of the total global macrofungus production (Table 1). The organoleptic properties of macrofungus and the quest for well-balanced diets led to a worldwide increase in the consumption of these materials (Wasser, 2011). Macrofungus production has been the sixth-largest

agricultural industry in China (Pan, Ai, Shao, Chen, & Gao, 2019). Moreover, macrofungus cultivation has also gradually increased globally (FAO, 2018). To obtain high-quality, cost-effective macrofungi, intensive processing has been required, especially in developing countries (Chang & Wasser, 2012).

As they may have strong bioactivity and pleasant and unique flavors, macrofungi serve as nutrient-rich foods, commercial nutraceuticals, and pharmaceutical products. The most widely cultivated edible macrofungi are *Lentinus edodes* (shiitake), *Pleurotus* spp (oyster macrofungi) (Bellettini et al., 2019), *Agaricus bisporus* (button macrofungi) (predominantly in the United Kingdom and other western countries) (Kabel, Jurak, Mäkelä, & de Vries, 2017), *Flammulina velutipes* (winter macrofungi) (Chang & Wasser, 2012), *Auricularia auricula* (wood ear macrofungi), and *Volvariella volvacea* (Zhang et al., 2013). Widely used medicinal macrofungi including *Ganoderma lucidum, Cordyceps sinensis* (Govorushko et al., 2019), *Phellinus linteus, Antrodia cinnaomea*, and *Xylaria nigripes* (Chang & Wasser, 2012) have been evaluated for their health-promoting properties (Friedman, 2016). However, with increasing environmental pollution, wild-crafted macrofungus harvest and artificial macrofungus cultivation with natural raw materials at risk of contamination by harmful metals such as Cd, Pb, Hg, Ag, and As, as well as radionuclides (Brzezicha-Cirocka, Grembecka, Grochowska, Falandysz, & Szefer, 2019; Falandysz & Borovička, 2013; H. Zou et al., 2019). Cooking significantly reduces the As levels in macrofungi (26% to 72%) but is relatively less effective at decreasing Hg, Cd, and Pb (Chiocchetti et al., 2020). Therefore, the development of alternate cultivation media and methods with specific safety ingredients may help circumvent this problem.

Submerged cultivation is a new technology that has been applied to macrofungi in recent years. It has numerous advantages over conventional solid culture such as a short incubation period, easy control of culture conditions, and ready availability of active metabolites and specific culture components. Additionally, submerged cultivation may increase the relative amounts of bioactive compounds that can be obtained from macrofungi (Rathore, Prasad, Kapri, Tiwari, & Sharma, 2019). Medium composition, culture temperature, altitude, luminosity, air composition, and especially exogenous additives may affect macrofungus yield and bioactivity. Therefore, appropriate production and cultivation conditions are vital to large-scale macrofungus cultivation, especially for rare macrofungi. To date, few reviews have summarized the factors influencing current macrofungi cultivation techniques.

The economic value of macrofungi may be enhanced by elucidating the relationships between the functions and bioactivity of their components. Numerous studies have shown that the bioactive constituents of macrofungi may prevent metabolic disorders and various diseases (De Silva et al., 2013; Liu, Jiao, Lu, Shu, & Chen, 2020; Roncero-Ramos & Delgado-Andrade, 2017). Thus, macrofungi extracts and their bioactive constituents could have extensive nutritional and therapeutic applications. Recently, several reliable studies demonstrated that certain macrofungi regulate gut microbiota. Hence, research investigations were made into the probiotic characteristics of macrofungi, which greatly expand the application of macrofungi. In general, macrofungi are applied towards developing innovative food and health care products with enhanced market value (Figure 2). However, few reviews have focused on current macrofungi applications or their corresponding mechanisms. The present review provides important insights into the cultivation, key bioactive components, and applications of macrofungi. As they possess numerous bioactive substances and their cultivation can be precisely controlled, macrofungi may become major natural health-promoting materials with high bioavailability.

2 | ARTIFICIAL MACROFUNGUS CULTIVATION

The bioactivity and structure–function relationships of the constituents in macrofungi have been clarified and reported. Nevertheless, only a few bioactive macrofungus-derived components have been commercialized. Relatively few studies focused on high-quality macrofungi because the bioactive constituents were unstable and expensive and their yields were low or erratic (Giavasis, 2014). Hence, practical high-yield macrofungus cultivation technology is urgently needed in order to meet market and consumer demands.

2.1 | Intrinsic factors

Most macrofungi are commercially cultivated by large-scale solid-state fermentation (SSF) that generates culinary or medicinal fruiting bodies and compensates for wood shortages in cut-log cultivation (Niksic, Klaus, & Argyropoulos, 2016). The substrates used in SSF macrofungus cultivation are mainly lignocellulosic wastes derived from wheat and rice straw, bran, rice hulls, forestry by-products, corn cobs, and others (Alananbeh, Bouqellah, & Al Kaff, 2014; Ramos et al., 2019; Yang, Guo, & Wan, 2013). Olive mill waste (Koutrotsios, Larou, Mountzouris, & Zervakis, 2016), coffee pulp, spent coffee grounds, and sisal waste are also used (Mata, Salmones, & Perez-Merlo, 2016). Unlike plants, macrofungi lack chlorophyll and are

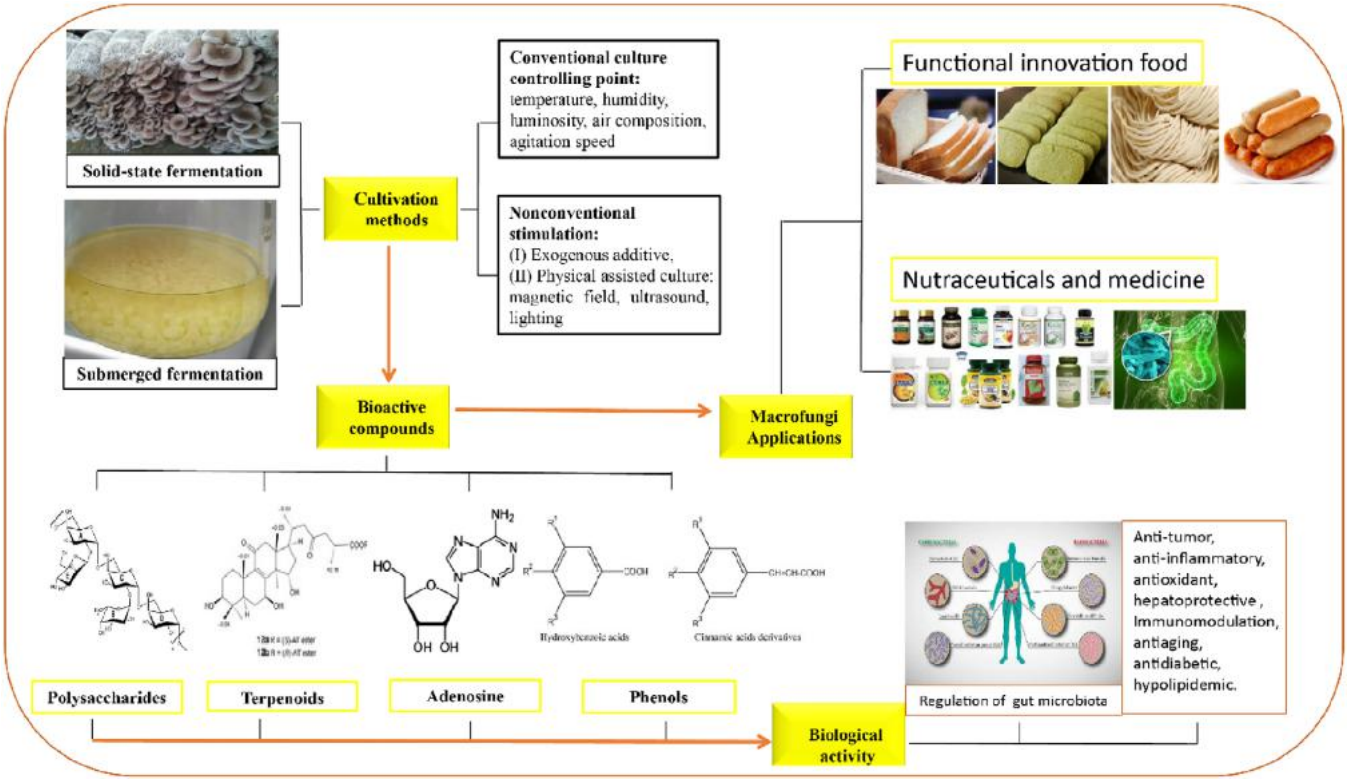


FIGURE 2 Comprehensive review of macrofungi cultivation, bioactive components, and applications

non-photosynthetic absorptive heterotrophs (Ramos et al., 2019). Macrofungi secrete enzymes to digest the lignocellulosic materials they absorb to support mycelial growth (Zhang et al., 2019). They also detoxify or bioconvert agroindustrial waste (Lakhtar, Ismaili-Alaoui, Philippoussis, Perraud-Gaime, & Roussos, 2010; Ntougias et al., 2012). In this way, they recycle agricultural residues (Sanchez, 2009). Various studies have indicated that synergistic biotransformations occurs between macrofungi and agroindustrial waste substrates (Koutrotsios & Zervakis, 2014; Koutrotsios et al., 2016). Macrofungi biodegrade and detoxify these materials by bioconverting them into high-quality, eco-friendly biomass (García-Delgado, Yunta, & Eymar, 2015). Moreover, the macrofungi derived from these substrates are highly nutritious and healthful.

The chemical compositions of wildcrafted and cultivated mushrooms differ (Reis, Barros, Martins, & Ferreira, 2012). Moreover, the nutritional value of cultivated macrofungi depends strongly upon substrate composition and cultivation technique (Koutrotsios & Zervakis, 2014; Liang, Wu, Lu, Kuo, & Liang, 2019). Thus, it is important to select suitable substrates for macrofungus cultivation. The amount of inoculum, moisture content, cultivation time, and substrate granularity must also be considered in SSF (Bellettini et al., 2019; Lou, Guo, Zhang, Guo, & Lin, 2019; Thomas, Larroche, & Pandey, 2013). Several recent studies showed that postharvest preservation and storage reduce browning and extend the shelf life of macrofungi in SSF (Manthou et al., 2019; Singh, Gaikwad, Lee, & Lee, 2018; Zhang, Pu, & Sun, 2018). However, these topics are beyond the scope of the present review. Whereas SSF is time consuming and labor intensive, submerged bioreactor fermentation is relatively rapid and easy and has other advantages over SSF. Submerged culture conditions and product quality can be conveniently controlled and microbial contamination can be mitigated (Elisashvili, 2012). Moreover, large quantities of mycelia and extracellular bioactive products can be generated (Rathore et al., 2019). The cultured mycelia are not directly used as food but may be utilized as food ingredients and flavoring agents (Rathore et al., 2019; Ulziijargal, Yang, Lin, Chen, & Mau, 2013). Several studies focused on the intrinsic factors of macrofungi that enable them to yield sufficient biomass and bioactive metabolites under optimized submerged fermentation conditions (Elisashvili, 2012; Tang, Zhang, & Zhong, 2009). Provided that there are sufficient numbers of high-quality spores, the substrate composition, pH, moisture, inoculation density, and carbon-to-nitrogen ratio (C/N) may dramatically influence mycelial biomass and bioactive products (Bellettini et al., 2019; Yang, Min, Bi, Zhou, & Huang, 2013). The constituents of the culture medium influence

TABLE 2 Effects of various exogenous additives on macrofungus mycelial growth

Macrofungus	Exogenous additive	Results	Reference
Pleurotus mutilis	20 g/L soybean oil	Mycelial BIO and pleuromutilin 49.10 g/L and 8.32 g/L, respectively	Hu, Zou, and Zhao (2009)
Ganoderma lucidum	2% coix lacryma-jobi oil	Mycelial BIO, triterpenoids, EPS, and IPS increased by 3.34-fold, 2.76-fold, 2.2-fold and 2.23-fold, respectively	Yang et al. (2013)
Ganoderma lucidum	2% corn oil	Mycelia BIO and polysaccharide 12.9 g/L and 1.038 g/L, respectively	Huang, Chen, Hung, and Liu (2009)
Grifola frondosa	1% plant oil: olive safflower seed, soy and sunflower oil 1% surfactant: Tween 80 and Span 80	Mycelial BIO and polysaccharides	Hsieh, Wang, Chen, Hsu, and Tseng (2008)
Phellinus ignarius L. Quél. *(Basidiomycota)*	4 g/L Tween-80	EPS yield and antioxidative activity 6.2 g/L and 5.5 mM, respectively	Ma et al. (2013)
Inonotus obliquus	1 g/L Tween 80	Biodegradation of ligninolytic enzymes to wheat straw and EPS antioxidant activity BIO and EPS increased by 23.3% and 142.9%, respectively	Xu et al. (2019)
Hericium erinaceum	5% *Ginseng* extract	Immunostimulation and anti-metastasis of mycelia	Kim et al. (2010)
Ganoderma lucidum	55 mg/L ethyl acetate extracts from *Eupolyphaga sinensis* and *Catharsius molossus*.	Mycelial BIO and IPS, EPS yields increased by 1.66-fold, 1.66-fold, and 1.36-fold, respectively	Liu and Zhang (2007)
Ganoderma lucidum	Methanol, ethanol, 1-propanol and 2-propanol at 1.5% (v/v)	Mycelial growth and polysaccharide	Yang, Wu, and Zhang (2004)
Pellinus linteus	Phytohormones: indole-3-acetic acid, indole-3-butyric acid, and 1-naphthaleneacetic acid at 1 mg/L, 1.5 mg/L, and 5 mg/L, respectively	Mycelial growth and EPS	Guo, Zou, and Sun (2009)

Abbreviations: BIO, biomass; EPS, exopolysaccharide; IPS, intracellular polysaccharide.

macrofungus safety. Allergic reactions to silkworm pupae were attributed to the presence of undissolved *Ophiocordyceps sinensis* culture residues (Choi, Shin, Kim, Ye, & Park, 2010). Thus, highly soluble culture media ensure comparatively safer cordyceps products and prevent the incorporation of residues that could lower product efficacy and cause side effects (Martel et al., 2017). The addition of exogenous additives such as surfactants, plant oils, organic solvents, fatty acids, phytohormones, and other precursors and chemical elicitors during macrofungus cultivation may promote mycelial growth and metabolite production (Table 2). Synergy between surfactants and plant oils in macrofungi has been reported (Bellettini et al., 2019; Postemsky, Delmastro, & Curvetto, 2014; Xu, Quan, & Shen, 2015). Surfactants may increase cell membrane permeability which, in turn, stimulates metabolite secretion and enzyme activity and the biosynthesis of secondary metabolites such as polysaccharides (Ma, Zhang, & Fam, 2013; Yang et al., 2013). Polysaccharides formation is affect by surfactants in several ways. Surfactants promote the transition from triple- to single-chain lentinans (Nandi, Guerra, Bellettini, Machado, & Minatti, 2013). They may also alter mycelial morphology, promote mycelial dispersion, and increase cell viability and product biosynthesis (Liu & Wu, 2012; Ma et al., 2013). Surfactants can also help macrofungi biodegrade wheat straw by upregulating ligninolytic enzymes such as Mn and lignin peroxidases, and laccase by 1,200%, 125%, and 39.9%,

respectively (Xu et al., 2019). However, the effects of most exogenous additives on macrofungi have not been well characterized. Few studies have elucidated the mechanisms by which exogenous additives upregulate genes or their protein products. Moreover, most of the exogenous additives previously investigated were inedible. The safety of exogenous additives in submerged macrofungus cultivation has not been systematically assessed. Determination of the effects of low-cost exogenous additives on high-value products derived from macrofungi may help optimize macrofungi cultivation. Moreover, the addition of biostimulants such as bacteria to *Coprinopsis cinerea* liquid fermentation or co-culture improved the yield and activity of antimicrobial peptides, proteins, sesquiterpenes, and novel fungal antibiotics. Biostimulants induce fungi to secrete various effectors at the synthase gene-encoding level (Essig et al., 2014; Kombrink et al., 2019; Stockli et al., 2019). Antagonistic signaling molecules such as bioactive secondary metabolites secreted in response to fungi-bacteria interactions may yield novel bioactive products.

2.2 | Extrinsic factors

Several extrinsic environmental factors and cultivation methods affect macrofungus quality. These may include culture temperature and heat treatment and culture house temperature used for substrate sterilization and mycelial growth, respectively. Other factors are humidity, luminosity, air composition, and agitation speed (Bellettini et al., 2019; Gurubel, 2017; Ren et al., 2019). Heat stress and different LED wavelengths in the *Cordyceps militaris* cultivation environment markedly stimulated fruiting body growth and cordycepin, carotenoid, adenosine, and exopolysaccharide formation (Chiang, Liang, Wang, & Liang, 2017; Jiaojiao et al., 2018; Kho et al., 2016). Heat- and light stress-induced genes encode a metal-dependent phosphohydrolase and an ATP phosphoribosyltransferase that participate in cordycepin biosynthesis (Jiaojiao et al., 2018).

An agitation speed of 200 rpm and an aeration rate of 2 vvm promoted cell mass and exopolysaccharide production in *Tremella fuciformis* cultured in a 5-L stirred tank and an airlift reactor (Cho, Oh, Chang, & Yun, 2006). Two-stage agitation speed control enhanced the triterpene production rate in *Ganoderma lucidum* by 49.22% (Feng et al., 2015). High pressure, electric fields, and ultrasound have been recently applied in microbial fermentation and could be used in macrofungus fermentation (Lu et al., 2020; Mota et al., 2018). Low-frequency magnetic fields and ultrasonic treatments affect mycelial growth and the biosynthesis of active substances (Gao, Zhang, & Feng, 2011; Zhang et al., 2014). They promote mass transfer and improve cell permeability by creating external and oxidative stress and inducing free radical formation. However, the culture device itself is also an extrinsic factor and may cause high shear stress and mycelial pellet aggregation. Conventional stirred tanks and airlift reactors cannot entirely correct for these factors (Domingos, Souza-Cruz, Ferraz, & Prata, 2017). Therefore, the effective design of a novel bioreactor is crucial in the quest for high-yield macrofungus cultivation.

Optimization of culture conditions is also important. Artificial neural networks and response surface methodology are used to optimize nutritional and environmental conditions for macrofungus cultivation. The quest for the enrichment and transformation of active ingredients and the enhancement of their shelf life has generated innovative physical technologies for the environmental macrofungus control including pulsed ultraviolet irradiation, γ-irradiation, and electron beam (Taofiq, Fernandes, Barros, Barreiro, & Ferreira, 2017). However, these techniques are generally unsuitable for the long cultivation periods required for macrofungus maturation and post-harvest.

3 | MACROFUNGUS INGREDIENTS AND BIOLOGICAL ACTIVITY

Macrofungi have antitumor, hypoglycemic, immunostimulatory, neuroprotective, anti-neuroinflammatory, cytotoxic, anti-hyperlipidemic, antiviral, hypotensive, antidiabetic, antioxidant, and antibacterial efficacy (Giavasis, 2014; Lau & Abdullah, 2017; Taofiq, Martins, Barreiro, & Ferreira, 2016; Tsai et al., 2009). These properties have been confirmed both in vitro and in vivo. The health-promoting effects of macrofungi are ascribed to their various bioactive metabolites. Macrofungi may play important roles in regulating the human gut microbiome (Chang et al., 2015; Giannenas et al., 2011; Li et al., 2019). This characteristic opens up new prospects for the application of macrofungi and their bioactive ingredients.

3.1 | Macrofungi active ingredients and biological activity

Certain macrofungi are good sources of essential amino acids, carbohydrates, and dietary fiber (Cohen et al., 2014; Reis et al., 2012; Sun, Zhang, & Fang, 2019). Some contain several nutraceutical compounds including bioactive polysaccharides, proteins, terpenoids, sterols, and antioxidants. Therefore, macrofungi can form integral parts of well-balanced diets and may be used in the clinical treatment of various degenerative diseases. Their bioactive constituents have additive and synergistic effects (Rathore et al., 2019; Xu, Yan, Tang, Chen, & Zhang, 2014).

TABLE 3 Active ingredients and health-promoting efficacy of commonly consumed macrofungi with various types of biological activity

Macrofungus	Active ingredients	Health-promoting function	Reference
Antrodia cinnamomea	Polysaccharides, triterpenoids, ubiquinone derivatives, succinic, and maleic derivatives	Antitumor, anti-inflammatory, antioxidant, hepatoprotective anti-fatigue, anti-obesity	Zhang et al. (2019)
Ganoderma lucidum	Polysaccharides, triterpenoids (mainly Ganoderma acid), proteins, steroids, sterols, nucleotides, fatty acids, vitamins	Immunomodulatory, anticancer, antidiabetic, antioxidant, anti-atherosclerotic, anti-fibrotic, chemopreventive, analgesic, anti-inflammatory, anti-nociceptive, antimicrobial, hypolipidemic, hepatoprotective, anti-androgenic, antiangiogenic, anti-herpetic, anti-arthritic, anti-osteoporotic, antiaging, antiulcer properties and estrogenic activity	Ahmad (2018)
Cordyceps sinensis,	Polysaccharides, proteins, sterols adenosine, cordycepin,	Antiaging, antioxidant, adaptogenic, aphrodisiac, neuroprotective, nootropic, immunomodulatory, and hepatoprotective	Chen, Wang, Fang, Dong, and Nie (2019); Chen, Wang, Nie, and Marcone (2013); Shashidhar, Giridhar, Udaya Sankar, and Manohar (2013)
Lentinus edodes (shiitake)	Lentinan, amino acids, vitamins (B1, B2, C, D), dietary fibers	Antitumor, antiaging, antioxidant, anti-hepatitis, immune-enhancing	Chung et al. (2019); Xiaofei Xu et al. (2014)
Pleurotus spp (oyster macrofungi)	Polysaccharides, peptides, amino acids, phenolics, terpenes, sterols, fatty acid esters,	Anti-inflammatory, immunomodulatory, anti-hypercholesterolemic, antihypertensive, antidiabetic, anti-obesity, antiaging, antimicrobial, antioxidant, hepatoprotective	Corrêa et al. (2016)
Agaricus bisporus (button macrofungi)	Polysaccharides, peptides, ergosterol vitamins (B2, niacin, and folates)	Immunomodulating, anticancer, antioxidant, hypoglycemic hypocholesterolemic, anti-inflammatory	Lau, Abdullah, Shuib, and Aminudin (2014); Ruthes et al. (2013)
Flammulina velutipes (winter macrofungi)	Polysaccharides bioactive protein, polypeptide, sterols, ergothioneine, cuparene sesquiterpenes, dilinolein, hemisceramide	Immunomodulatory, anti-tumor, antioxidant, hepatoprotection, anti-hyperlipidemic	Cai, Liu, Chen, Liao, and Zou (2013); Zhang et al. (2013)
Auricularia auricular (wood ear macrofungi)	Polysaccharides, melanin, vitamins	Antimicrobial, antioxidant, antitumor, anticoagulant, hypotensive and antilipidemic activities	Wu, Zhang, Yang, Zhou, and Yang (2018)

The fruiting bodies, mycelia, and fermentation broths of macrofungi furnish numerous restorative compounds with immunomodulatory and anticancer efficacy such as β-glucans, lectins, triterpenoids, tocopherols (Cordara, Bjerregaard-Andersen, & Krengel, 2018; Roncero-Ramos & Delgado-Andrade, 2017), low-molecular-weight (LMW) compounds (terpenes, steroids, anthraquinones, benzoic acid derivatives, and quinolones), and high-molecular-weight (HMW) compounds (peptides and proteins) with antimicrobial efficacy (Reis, Martins, Vasconcelos, Morales, & Ferreira, 2017). The bioactive ingredients and health-promoting efficacy of common bioactive constituents in macrofungi are listed in Table 3. Certain macrofungi also contain medicinally bioactive

ingredients such as lovastatin, γ-aminobutyric acid (GABA), and L-ergothioneine with cholesterol-lowering, anti-neuroinflammatory, hypotensive, antioxidant, and cytoprotective properties (Chen, Ho, Hsieh, Wang, & Mau, 2012; Cohen et al., 2014).

Polysaccharides, saponins, triterpenes, adenosines, and flavonoids are major categories of macrofungal components approved for use in health foods (Li, Ma, & Zhang, 2019; Sun et al., 2019). The polysaccharides, terpenoids, and phenolic compounds in the fruiting bodies, mycelia, and culture broth of *Antrodia camphorata, Agaricus bisporus, Cordyceps sinensis, Pleurotus* spp., and *Ganoderma lucidum* (Corrêa, Brugnari, Bracht, Peralta, & Ferreira, 2016; Taofiq et al., 2016; Wu et al., 2017), have potent anti-inflammatory efficacy (El Enshasy & Hatti-Kaul, 2013; Taofiq et al., 2016; Wu et al., 2017). They can be used to treat conditions associated with inflammation such as diabetes, arthritis, obesity, metabolic syndrome, cancers, and cardiovascular diseases (Du, Zhu, & Xu, 2018; Taofiq et al., 2016). A recent study showed that phenolics such as chlorogenic and syringic acids extracted from macrofungi are antiangiogenic and restrict blood vessel development by inhibiting vascular endothelial growth factor (VEGF) (Lin, Ching, Chen, & Cheung, 2015).

Bioactive macrofungal polysaccharides (MPs) are often described as "biological response modifier." Numerous studies have shown that they have antioxidant, anti-inflammatory, anticancer, antidiabetic, antimicrobial, antilipidemic, anti-hypoxic, hypoglycemic, and immunomodulatory activity (Corrêa et al., 2016; Jiao, Kuang, Wu, & Chen, 2019; Liu, Choi, Li, & Cheung, 2018; Yan et al., 2019). Most MP functions and bioactivity are related to immune system function (Shi, 2016). MPs are β-glucan polymers whose main chains consist of β-(1 → 3) linkages with β-(1 → 6) branches (Friedman, 2016). The biological properties of MPs depend on their structural and physicochemical characteristics such as MW, chain conformation, presence of double or triple helices or random coils, degree of branching, non-sugar components (protein complexes or sulfate groups), and extraction methods (Ioannis Giavasis, 2014; Meng, Liang, & Luo, 2016; Shi, 2016). MP activity is affected by the bioactivities of the extraction solvent and other components in the food (Giavasis, 2014; Sun et al., 2019). The solvent used to prepare the macrofungi extract may determine whether the product has an immunosuppressive or immunostimulant effect. Aqueous extracts of *Agaricus blazei* Murrill, *Antrodia cinnamomea, Ganoderma lucidum*, and *Ophiocordyceps sinensis* activated immune cells whereas ethanolic extracts of these same macrofungi inhibited them (Lu et al., 2016; Martel et al., 2017).

Well-characterized macrofungal polysaccharides include lentinan, schizophyllan, krestin, ganoderan, and β-glucans from *Agaricus blazei*, *Grifola frondosa*, *Trametes versicolor*, *Pleurotus ostreatus*, and others (Giavasis, 2014; Rathore et al., 2017). Grifolan, lentinan, krestin, and schizophyllan have been used in clinical medicine and are available on the market (Ioannis Giavasis, 2014; Liu et al., 2018). MPs comprise homopolysaccharides, heteropolysaccharides (polysaccharide-protein complexes), and chitin (Ramos et al., 2019; Zhang et al., 2011). The most important therapeutic component of macrofungal polysaccharides is β-glucan (homopolysaccharide) (Friedman, 2016; Kothari, Patel, & Kim, 2018). Chitin and β-glucan are major components of macrofungal cell walls, have a wide range of physiological functionalities (Bai et al., 2019), and have been extensively investigated (Roupas, Keogh, Noakes, Margetts, & Taylor, 2012). Heteropolysaccharides also possess biological properties (Datta et al., 2019). Macrofungus-derived β-glucan administration has immunostimulant and antitumor activities (Du, Lin, Bian, & Xu, 2015; Rathore et al., 2017). β-Glucan modulates adaptive and innate immunity (Figure 3) and has indirect antitumor efficacy (Meng et al., 2016). Polysaccharides or β-glucan derived from macrofungi can directly or indirectly induce cancer cell apoptosis (Lee & Hong, 2010; OuYang et al., 2013; Meng Zhang, Zhang, Zhang, & Tian, 2019), inhibit tumor angiogenesis (Meng et al., 2016; Sze & Chan, 2012), and arrest cell cycle (OuYang et al., 2013). Lentinan has been approved as a clinical adjuvant therapy for cancer. It is used mainly to treat lung, gastric, colorectal, ovarian, cervical, pancreatic, cardiac, and nasopharyngeal cancers as well as non-Hodgkin's lymphoma (NHL) (Zhang et al., 2019). As a result of its characteristic and unique water solubility, viscosity, and gelation (Zhu, Du, & Xu, 2016), β-glucan is used as a thickener in the food industry and other elsewhere (Barone Lumaga, Azzali, Fogliano, Scalfi, & Vitaglione, 2012).

3.2 | Macrofungi regulate gut microbiota

Several investigations reported that certain indigestible macromolecules in macrofungi might function as prebiotics that stimulate the growth of gut microbiota (Delzenne & Bindels, 2015; Holmes, 2015; Parnell & Reimer, 2012; Wu et al., 2019). The gut microbiome comprises trillions of bacteria, viruses, fungi, archaea, bacteriophages, and protozoans. It regulates energy acquisition, prevents intestinal pathogen infection, and modulates normal immunological responses (Chang et al., 2015). Hence, it is closely associated with human metabolic health (Dethlefsen, McFall-Ngai, & Relman, 2007; Karlsson et al., 2013; Tremaroli & Backhed, 2012) and the expression of genes encoding certain disorders (Schele et al., 2013), obesity (David Holmes,

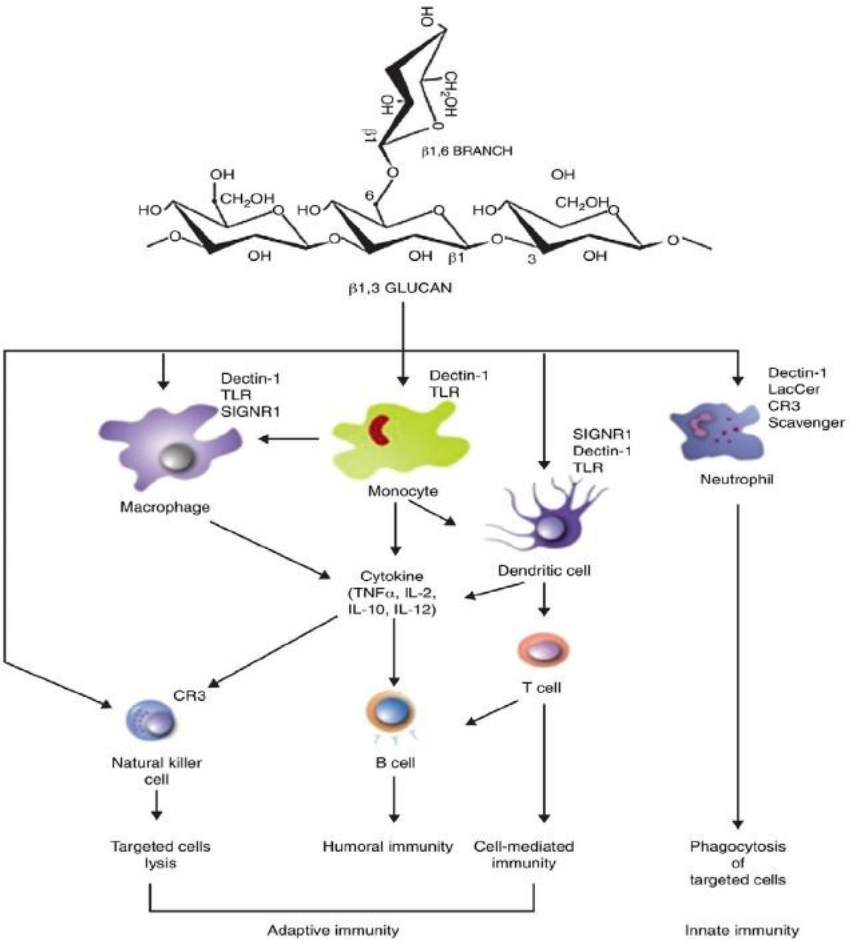


FIGURE 3 Immunoprotective mechanisms of macrofungal β-glucans. β-glucans modulate innate and adaptive immune responses by binding several immune cell membrane receptors such as dectin and Toll-like receptor. In this manner, they activate multiple signaling pathways involving macrophages, monocytes, neutrophils, and dendritic, natural killer, B, and T cells. β-glucans stimulate the release of cytokines such as TNF-α and several interleukins (Ioannis Giavasis, 2014). Abbreviations: TNF-α, tumor necrosis factor-α. ILs, interleukins

2015; Liu et al., 2017; Ridaura et al., 2013), type 2 diabetes (Gu et al., 2017; Qin et al., 2012), cardiovascular disease (Jie et al., 2017), inflammatory and autoimmune diseases, neurological conditions, metabolic endotoxemia, and cancers (Duvallet, Gibbons, Gurry, Irizarry, & Alm, 2017; Jackson et al., 2018). The gut microbiota community of obese mice and humans had higher ratios of *Firmicutes* to *Bacteroidetes* than those of their lean counterparts (Zhao, 2013). Gut-related conditions such as obesity, diabetes, and inflammatory bowel diseases (IBD) have been extensively investigated.

Diet is a key factor regulating gut microbiota and may, therefore, influence host metabolism (Dalby, Ross, Walker, & Morgan, 2017). A randomized open-label crossover study was conducted on healthy adults consuming protein-matched quantities of either *Agaricus bisporus* or meat twice daily for 10 days. Both treatments were generally well-tolerated. The mushroom-rich diet resulted in relatively higher average stool weight and different fecal microbiota composition including a higher abundance of *Bacteroidetes* and a lower abundance of *Firmicutes* (Hess, Wang, Gould, & Slavin, 2018). Introduction of probiotics, prebiotics, or synbiotics into the human diet may balance the host intestinal ecosystem by improving gut microbiota integrity and reducing the incidence and severity of metabolic disorders (Markowiak & Slizewska, 2017; Yoo & Kim, 2016). Polysaccharides and dietary fiber play key roles as prebiotics and are candidates for the development of antidiabetic and anti-obesogenic treatments (Chang et al., 2015; Gudi et al., 2019).

Moreover, macrofungual extracts could be used as prebiotics to prevent or treat hyperlipidemia, hypercholesterolemia, obesity-related metabolic disorders, inflammation, and type 2 diabetes by regulating gut microbiota abundance and maintaining intestinal barrier integrity (Table 4).

Macrofungi are abundant in indigestible polysaccharides such as chitin, hemicellulose, β-glucan, α-glucan, mannans, xylans, galactans, and polysaccharide-peptide/protein complexes (Friedman, 2016; Singdevsachan et al., 2016). Ingested MPs are not readily hydrolyzed by human or animal pancreatic digestive enzymes but they are absorbed in the upper gastrointestinal tract (Delzenne & Bindels, 2015). After they enter the large intestine, they are decomposed into short-chain fatty acids (SCFAs) and gases by carbohydrate-active

TABLE 4 Macrofungus and their extracts treat metabolic disorders by regulating gut microbiota

Macrofungi	Extract type	Bioactive compound	Models	Outcomes	Mechanisms	References
Ganoderma lucidum	Water extract of mycelium	HMW polysaccharides	HFD-fed mice daily by oral gavage for 2 mo	Reduce weight and fat, inflammation and insulin resistance	Reduce the ratio of *Firmicutes* to *Bacteroidetes* and levels of *Proteobacteria*	(Chang et al., 2015; Holmes, 2015)
Ganoderma lucidum	Water extract of fruiting body	Polysaccharide peptides	HFD-fed mice fed for 8 wks	Ameliorate lipid metabolism disorders	Phylotypes (-) Jeotgalicoccus (-), Ignavigranum (-), *Sporosarcina* (-), *Bacteroides* (-), *Anaerovorax* (-), *Parasutterella* (-), *Alistipes* (-) *Alloprevotella* (-), *Allobaculum* (+), *Phascolarctobacterium* (+), *Psychrobacter* (+)*Enterorhabdus* (+), *Blautia* (+) *Roseburia* (+)	(Lv, Guo, Li, Yu, & Liu, 2019)
Ganoderma lucidum	55% ethanol extracts	Enriched with triterpenoids	HFD-fed mice	Ameliorate lipid metabolism disorders and alleviate dyslipidemia	*Alloprevotella* (+) *Ruminococcus* (+) *Prevotella* (+), *Turicibacter* (-) *Clostridium* XVIII (-)	(Hu et al., 2018)
Hirsutella sinensis	Microwave-assisted water extraction	HMW-polysaccharides (> 300 kDa)	HFD-fed mice	Reduced intestinal and systemic inflammation. Improved insulin sensitivity and lipid metabolism	*Parabacteroides goldsteinii* (+)	(Wu et al., 2019)
Antrodia cinnamomea	Water extract of fruiting body	Not clear	Male C57BL/6J mice fed HFD for 8 wks	Reduce insulin resistance, inflammation, weight, fat, and blood endotoxemia	Lower *Firmicutes*/*Bacteroidetes* ratio and *Akkermansia* (+) *muciniphila* (+)	(Chang et al., 2018)

(Continues)

TABLE 4 (Continued)

Macrofungi	Extract type	Bioactive compound	Models	Outcomes	Mechanisms	References
Grifola frondosa	Water extract of fruiting body	HMW-polysaccharides	HFD-fed mice 8 wks	Prevent bodyweight gain and ameliorate lipid metabolism disorders	*Clostridium*-XVIII (+), *Butyricicoccus* (+) *Turicibacter* (+), *Helicobater* (-), *Intestinimonas* (-), *Barnesiella* (-), *Parasutterella* (-), *Ruminococcus* (-) *Flavonifracter* (-).	(Li et al., 2019)
Grifola frondosa	Water extract of fruiting body	GFP-N, an HMW-heteropolysac-charide, (12,600 k Da)	type 2 diabetic male ICR mice	Activate insulin receptor, phosphatidylinositol-3-kinase, and glucose transporter for hypoglycemic effect	*Bacteroidetes* (+) *Akkermansia* (+), *Lactobacillus* (+), *Turicibacter* (+) *Firmicutes* (-) *Proteobacteria* (-)	(Chen et al., 2019)
Flammulina velutipes	Water extract of freeze-dried powder	HMW-polysaccharides with β- glycosidic linkages	Six-week male C57BL/6 mice	Suppression of neuroinflammation and improving learning and memory impairment	*Clostridia* (-) *Bacilli* (-) *Bacteroidia* (+), *Erysipelotrichia* (+) *Actinobacteria* (+)	(Su et al., 2018)
Flammulina velutipes	Water extract of fruiting body	Crude polysaccharide with β-glycosidic bonds	Administered to 6-wks male ICR mice	Increase SCFAs, immuno-globulins and immune factors, weight reduction	Lower *Firmi-cutes*/*Bacteroidetes* ratio and increase fecal microbial diversity	(Zhao et al., 2019)

(Continues)

TABLE 4 (Continued)

Macrofungi	Extract type	Bioactive compound	Models	Outcomes	Mechanisms	References
Hericium erinaceus	Water extract of mycelium powder	Polysaccharide of *H. erinaceus* (MW: 86.67 kDa)	C57BL/6 mice	Reduce IBD and down-regulate key markers of oxidative stress	Restore relative abundances of vital bacteria including *Clostridiales*, *Akkermansia*, and *Desulfovibrio*	(Ren et al., 2018)
Agaricus bisporus	Freeze-dried powder	Not clear	C57BL/6 mice	Prebiotic effects, produce succinate/propionate, improve glucose homeostasis via intestinal gluconeo-genesis	Expand *Prevotella* population	(Tian et al., 2018)
Auricularia auricular	Water extract of freeze-dried powder	Polysaccharides with β-type glycosidic bonds (mostly HMW)	Male ICR mice	Increase SCFAs and immune-globulin	Decrease *Firmicutes*/*Bacter-oidetes* ratio *Porphyro-monadaceae*(+). *Bacteroidaceae* (+)	(Zhao et al., 2019)
Pleurotus eryngii	Water extract of freeze-dried powder	HMW-polysaccharides (426 kDa) with β-glycosidic linkages	Male C57BL/6 mice (6 wks)	Increase SCFA and alter immune response	*Porphyro-monadaceae* (+), *Rikenellaceae* (+), *Bacteroidaceae* (+) *Lactobacillaceae* (+)	(Ma et al., 2017)

(+), increase in abundance; (-), decrease in abundance. Abbreviations: HMW, high-molecular-weight; HFD, high-fat diet; SCFAs, short-chain fatty acids.

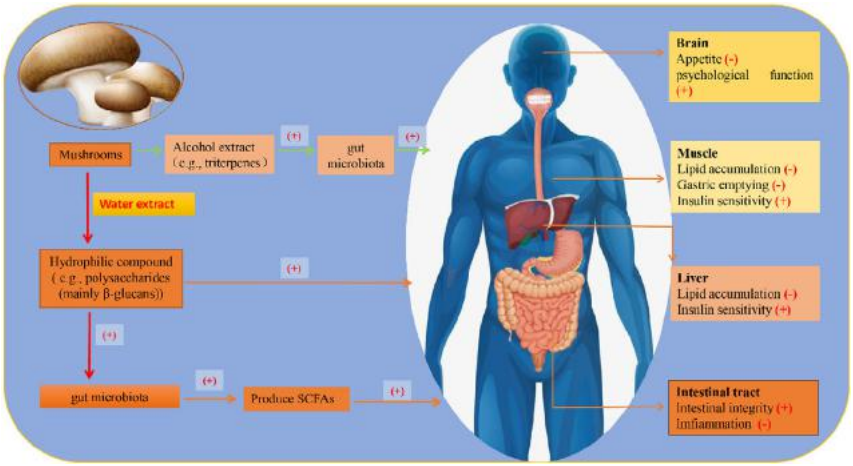


FIGURE 4 Bioactive macrofungal constituents may be degraded by gut microbiota into SCFAs that are beneficial to host health. (+), promotion; (-); inhibition. Abbreviation: SCFAs, short-chain fatty acids

enzymes (CAZymes) secreted from gut microbiota such as *Bacteroides* spp. In turn, the digestion products induce *Bifidobacterium* spp. which help promote and maintain colon health (Canfora, Jocken, & Blaak, 2015; Chang et al., 2015; Rodrigues et al., 2016). SCFAs such as acetate, propionate, butyrate, and valerate are essential nutrients for the colonic epithelium. They participate in microbiota–gut–brain crosstalk and may influence immune, antitumor, endocrine, neural, and humoral functions as well as appetite and mental state. All of these are beneficial to host health (Chang et al., 2015; Dalile, Van Oudenhove, Vervliet, & Verbeke, 2019; Rowland et al., 2018) (Figure 4). MPs upregulate occludin and zonula occludens protein 1 (ZO-1) in the intestinal epithelium. These substances control epithelial cell permeability and intestinal mucosa integrity. MPs also prevent the release of proinflammatory bacterial endotoxins into the bloodstream (Martel et al., 2016).

Hence, MP intake may modulate gut microbiota abundance, reduce bacterial endotoxin levels, and restore imbalances in gut microbiota caused by obesity, diabetes, and other conditions. A recent study demonstrated that in obesity, supplementation with MPs from *Ganoderma lucidum* (>300 kDa) improved gut barrier integrity, reduced endotoxemia, decreased TLR4 signaling and inflammation, diminished macrophage infiltration, and enhanced regulatory T cell accumulation in hepatic and adipose tissues. The beneficial effects of MPs may be attributed to the specific alterations they cause in gut microbiota such as lowering *Escherichia* spp. and *E. fergusonii* titers and maintaining gut barrier integrity (Chang et al., 2015). Several polysaccharides varying in size may contribute to the favorable effects of MPs against certain diseases (Delzenne & Bindels, 2015). Wu et al. (2019) reported that HMW polysaccharides (>300 kDa) from *Hirsutella sinensis* reversed obesity-induced gut dysbiosis and leaky gut and, by extension, markedly reduced body weight gain and metabolic disorders such as endotoxemia, inflammation, insulin resistance, and dyslipidemia. They could serve as novel prebiotics that selectively promote the growth of the commensal gut bacterium *Parabacteroides goldsteinii* for the treatment of obesity and type 2 diabetes. Macrofungal proteins extracted from *Hericium erinaceus* have immunomodulatory activity. They induce T cell proliferation and differentiation by regulating gut microbiota composition and metabolism. Macrofungal proteins are nitrogen sources and growth factors that significantly promote the engraftment of bifidobacteria and other probiotics (Diling et al., 2017). Therefore, Macrofungal proteins may serve a prebiotic function in inflammatory bowel diseases (IBD) associated with excessive antibiotic administration. Nevertheless, the mechanisms by which macrofungal proteins regulate gut microbiota remain to be elucidated. Certain studies reported that protein-bound glucan (PBG) and other polysaccharide-protein combinations from the edible mushroom *Coriolus versicolor* are potent anti-obesity factors. They upregulate several genes associated with host metabolism, alter host physiology, and create an intestinal microenvironment conducive to *Akkermansia muciniphila* colonization (Li et al., 2019). Research on macrofungus-derived probiotics and prebiotics may also contribute to animal nutrition and health (Giannenas et al., 2011; Solano-Aguilar et al., 2018). The bioactive substances of macrofungi that regulate gut microbiota are mainly HMW polysaccharides and traces of polysaccharide peptides. However, it remains to be determined whether macrofungal ingredients other than polysaccharides also benefit host health by influencing gut microbiota. Other macrofungal components such as sugar, glycoprotein, and glycolipid complexes and non-polysaccharide compounds including proteins, triterpenes, and polyphenols should also be systematically evaluated to determine their roles in shaping gut microbiota.

TABLE 5 Macrofungus-based food products

Macrofungus	Food products	Effect and result	References
Mycelia of *Antrodia camphorata*, *Agaricus blazei*, *Hericium erinaceus*, and *Phellinus linteus*	Bread	Higher umami intensity; maintenance of substantial amounts of GABA and ergothioneine	(Ulziijargal et al., 2013)
Pleurotus sajor-caju fruiting body	Biscuit	Increased dietary fiber, β-glucan and protein content of biscuit and improve postprandial glycemic response	(Ng et al., 2017)
Lentinus edodes	Rice noodle	Strengthen rice noodle extensibility and firmness	(Heo, Jeon, & Lee, 2014)
Pleurotus sapidus	Vegan sausage	Alternative to commercial vegetable proteins	(Stephan et al., 2018)
Grifola frondosa, *Ganoderma lucidum*	Fermented tea	Improve sensory flavor and therapeutic qualities of fermented tea	(Terrien, 2017)

4 | MACROFUNGAL FOOD PRODUCT DEVELOPMENT

Macrofungal mycelia and fruiting bodies contain multiple nutrients and bioactive health components such as amino acids, proteins, dietary fibers, and active polysaccharides as well as functional components such as lovastatin, gamma-aminobutyric acid (GABA), and ergothioneine. Thus, they are good sources of healthy food, nutraceuticals, and food-flavoring additives (Ulziijargal et al., 2013). At this time, novel applications for mycelia- and fruiting body-based macrofungal foods are being explored in the improvement of food flavor and nutrition. This approach might increase the industrial value of macrofungi without reducing product acceptability (Rathore et al., 2019) (Table 5). Nowadays, macrofungi are used mainly in flour-based products such as breads and biscuits. They enhance the antioxidant and nutritional content of pasta, decrease the digestibility of its starch, and lower the glycemic response it induces (Lu et al., 2018). A study conducted on *Pleurotus sajor-caju* powder disclosed that it could alter starch digestibility and granule structure in biscuits and improve postprandial glycemic response (Ng, Robert, Wan Ahmad, & Wan Ishak, 2017). When *Agaricus bisporus* polysaccharide was added to gluten-free flours, it had positive effects on the functional, pasting, rheological, and sensory properties of flour and biscuit doughs and made them fit for consumption by celiac patients (Sulieman et al., 2019). Macrofungus-derived proteins may substitute for animal proteins in vegan sausage production (Stephan, Ahlborn, Zajul, & Zorn, 2018; Terrien, 2017) and serve as meat analogs (Kim et al., 2011). Macrofungal protein has higher acceptability than soybean protein. The use of macrofungi as meat substitutes has been investigated for the purpose of producing healthy functional foods and diminishing the environmental footprint of animal husbandry. Macrofungus powder may ameliorate the rheological and structural characteristics of meat emulsions and help formulate functional products with superior sensorial and other characteristics (Kurt & Genccelep, 2018). Macrofungus powder might replace the phosphates used in emulsion-type sausages to minimize fat separation and lipid peroxidation and improve product flavor, taste, and acceptability (Choe et al., 2018; Pil-Nam et al., 2015). Macrofungi are also being exploited to develop novel tea beverages and fermented rice foods (Terrien, 2017). In practice, however, there are comparatively few studies on the development of innovative macrofungus-based foods. Macrofungi are used in the production of kombucha (Sknepnek et al., 2018), beer (Leskosek-Cukalovic et al., 2010), and health wine.

Macrofungi are rich in numerous bioactive ingredients and metabolites. Nevertheless, extracting these components from macrofungi for application in food ingredients requires further investigation. The type of extraction method selected will affect the structure and yield of these bioactive ingredients, thereby influencing their biological activity. Therefore, it is necessary to establish a practical and effective extraction method that ensure the high-yield of efficacious bioactive substances. Considerable attention has been directed toward the application of MPs in functional foods and health products. The most bioactive polysaccharides in macrofungi have complex structures and are indigestible in the host gastrointestinal tract. Hence, indigestible macromolecular MPs could serve as beneficial prebiotics for gut microbiota in the host intestine. Prebiotics modulate colonic microbiota composition, lower pH, increase SCFA production, and stimulate colonic immune responses in the colon (Aida, Shuhaimi, Yazid, & Maaruf, 2009; Chang et al., 2015; Delzenne & Bindels, 2015; Zhao, Cheng, Nakata, Zhao, & Hu, 2019; Zhao, Yang, Pei, Zhao, & Hu, 2018). In certain cases, macrofungal prebiotics could function as cost-effective substitutes for generally accepted and applied carbohydrate polymer and oligomer prebiotics such as fructooligosaccharides (FOS), galactooligosaccharides

(GOS), and inulin (Aida et al., 2009). As a result of their particular physicochemical properties, MPs also serve as and gelation agents in aqueous media and form biofilms conducive to probiotic survival and antibiotic activity (Liu, Wang, Zhao, & Wang, 2013; Mao, Song, Yao, & Wu, 2018). Polysaccharides from the stipes of *Lentinula edodes* and the bases of *Pleurotus eryngii* and *Flammulina velutipes* prolonged probiotic survival during cold storage and in simulated gastric and bile juices. Hence, these materials could be used in yogurt (Chou, Sheih, & Fang, 2013). As macrofungi are rich in umami components, they have appealing flavors and are ideal food flavoring agents. Nonvolatile and volatile compounds in shiitake extract could enhance cooked minced meat flavor (Sknepnek et al., 2018). The phenolic compounds in macrofungi have strong antioxidant and bacteriostatic activity and could, therefore, be applied in the food industry as natural antioxidants and antimicrobial agents (Bach et al., 2019).

5 | SAFETY CONCERNS OF MACROFUNGUS CULTIVATION AND CONSUMPTION

As they are rich in a multitude of bioactive components, macrofungi play important roles in food, health care products, as well as cosmetics. The potential toxicities, if any, are already known for most edible macrofungi. β-Glucan from *G. lucidum* has been used as a dietary supplement and food ingredient. It have not induced any significant subchronic toxicity in rodents and was not mutagenic (Chen et al., 2011). These findings were consistent with those of subchronic toxicity and genotoxicity studies on β-glucan from *Antrodia cinnamomea* (Chen, Chang, Chen, & Soni, 2018). The no observable adverse effects level (NOAEL) for mushroom β-glucan was 2,000 mg/kg BW/day. Though macrofungi abound in various active ingredients, only a few reports have indicated that edible macrofungi caused any negative effects in vivo (Rebolj et al., 2007). A lethal protein in *Agrocybe aegerita* was linked to hepatotoxicity (Jin et al., 2014). Sporadic intoxication events in humans and animals after the ingestion of large quantities of fresh macrofungi have been reported (Zuzek, Macek, Sepcic, Cestnik, & Frangez, 2006). Thermolabile proteinaceous molecules in macrofungi may contaminate the culture environment as well as the edible macrofungal products themselves (Zuzek et al., 2006). The adverse effects of the toxins in certain edible macrofungi on human health remain controversial. It was reported that the ingestion of yellow knight mushrooms (*Tricholoma equestre*) could cause rhabdomyolysis (Bedry et al., 2001). After extensive and numerous human trials on *T. equestre* intake, however, no hematological or biochemical alterations and no other adverse effects were observed (Klimaszyk & Rzymski, 2018). Therefore, macrofungi toxicity and safety merit consideration. Systematic safety assessments should be conducted on macrofungi and the substances extracted from them. Concerns about food quality control and safety have increased along with the demand for macrofungi. To improve macrofungi acceptability, it is necessary to evaluate their safety and toxicity.

Macrofungi growing in polluted areas accumulate heavy metals by absorbing them from the ambient environment (Melgar, Alonso, & García, 2016). Macrofungi may also accumulate radionuclides (Chiocchetti et al., 2020; Zou et al., 2019). In mycoremediation, macrofungi act as bio-sorbents and remove toxic metals from soil and wastewater. Macrofungi can also serve as biomarkers to assess environmental pollution levels. *Pleurotus ostreatus* decontaminated heavy metal pollution in colliery effluent (Vaseem, Singh, & Singh, 2017). The secondary metabolites of macrofungi may also adsorb harmful materials. Alkali-soluble polysaccharides isolated from *Boletus edulis* have strong binding affinities for Cd and Pb (Choma et al., 2018). Therefore, the endogenous toxins and environmental contaminants in edible macrofungi constitute serious food safety risks. Consequently, macrofungus cultivation on contaminated substrates and wildcrafted macrofungus harvest have constrained the development of the macrofungus industry. Cooking and digestion do not entirely eliminate heavy metal toxicity in macrofungi. As the bioactive constituents of macrofungi continue to be explored, the safety of macrofungi and their extracts must be carefully assessed. Heavy metal contamination in macrofungi raised in cultivation areas must be thoroughly investigated. Degradation of heavy metals by cooking and digestion should also be evaluated. Metabolite bioaccumulation in macrofungi in response to toxic and radioactive elements should be examined. Edible macrofungi may also be transmission routes of intracellular foodborne pathogens such as *E. coli* O157:H7, *Listeria monocytogenes*, *Salmonella* spp., *Staphylococcus aureus*, and *Yersinia enterocolitica*. For 402 samples of 22 species of cultivated and wild fresh mushrooms sold in retail markets, the microbial counts were in the range of 4.4 to 9.4 log CFU/g (Venturini, Reyes, Rivera, Oria, & Blanco, 2011). Approximately 1.20% of 665 edible macrofungus samples tested positive for hypervirulent *L. monocytogenes* strains (Chen et al., 2018). As a result, microbiological contamination and safety must be considered in macrofungus processing and consumption.

6 | FUTURE PERSPECTIVES

In order to cultivate edible fungi stably and efficiently, genetic engineering could be applied to develop stable

strains with commercially desirable traits and genes controlling mycelial primordium formation. In this way, macrofungi may be designed and bred for large-scale, high-quality production. To assure product authenticity, purity, quality, and safety, correct macrofungus species identification must be confirmed by phylogenetic trees, multigene co-sequencing by 28S nrDNA and 18S nrDNA, ITS, IGS, RNA polymerase II second largest subunit (*rpb2*), translation elongation factor 1-alpha gene (*tef1*), β-tubulin (*tub2*), DNA fingerprints, genome-wide microsatellites, and comprehensive microsatellite marker databases. Morphological comparisons alone do not suffice for this purpose as functionally different macrofungi may nonetheless physically resemble each other. Multiple sequence alignments and phylogenetic analyses should be developed to guide and regulate product synthesis. Biosynthesis of macrofungal substances could be genetically manipulated by controlling the expression of the key genes governing bioactive constituent biosynthesis. Therefore, regulation of the target receptors of the gene products controlling macrofungal growth and secondary metabolism must be explored. In this way, novel macrofungal-derived products can be biosynthesized.

The interactions between macrofungal components and gut microbiota are still unknown. Gut microbiota may transform bioactive phytochemicals by hydrolysis, oxidation, reduction, isomerization, rearrangement, esterification, condensation, and other reaction mechanisms (Feng, Ao, Peng, & Yan, 2019). Nevertheless, the effects of gut microbiota on the macrofungual biotransformation and the impacts of the reaction products on human health have not been established. Several studies focused on the impacts of macrofungal component ingestion on gut microbiota diversity and abundance. In the future, these effects should be compared with those of known prebiotics such as FOS, GOS, and inulin already used to treat gastrointestinal diseases. Mimicking human microbial metabolism by germ-free humanized gnotobiotic mice, human-microbiota associated (HMA) animals can provide a reliable model to study the human gut microbial ecosystem and metabolism. Advances in next-generation (DNA) sequencing (NGS), metagenomics, metatranscriptomics, metaproteomics, and metabolomics can identify functional links between metabolism and human gut microbiota.

Macrofungi have numerous biological activities. However, neither the efficacy nor the safety of their constituents has been validated at the clinical trial level. Thus, most of the bioactive ingredients in macrofungi are classified as health care products rather than drugs for the adjunctive treatment of various diseases. The functions of the main bioactive ingredients in macrofungi via targeted or untargeted metabolomics, elucidation of their molecular structures, and in vivo and in vitro bioactivity assays. In this manner, the scope of their application can be expanded. The relationships between the bioactive ingredients and their modes of action should be determined at the molecular level by multi-omics approaches. The bioactivity of other newly identified components may be predicted by quantitative structure-activity relationships (QSAR) and pharmacophore-based virtual screening. Moreover, normally consumed quantities of edible macrofungi have no known toxicity. Nevertheless, safety of the bioactive ingredient in macrofungi targeted for disease prevention or treatment must be evaluated by food and candidate drug toxicology assays such as the Ames, in vitro chromosome aberration, acute oral toxicity, and bone marrow micronucleus tests. For this reason, standard protocols for the prophylactic or therapeutic administration of dietary supplements must be established and implemented at the clinical stage level.

7 | CONCLUDING REMARKS

Macrofungi have great potential research value and varied application prospects as they are rich in metabolites and other bioactive constituents. Currently, macrofungi are mainly consumed either fresh or dried under mild processing conditions. The development of highly processed macrofungal derivatives such as condiments and ready-to-eat foods and the expansion of the industry chain may add considerable value to macrofungi. Exploitation of the bioactive ingredients in macrofungi may dramatically increase the market value. Therefore, applying macrofungi and their useful derivatives is important. Furthermore, novel systems for the mass production of macrofungi and their active substances merit design and implementation. Moreover, further research is required to elucidate primordia formation by macrofungal fruiting bodies, spore germination, and mycelium formation in liquid fermentation mycelia. In addition, the influences of circadian rhythm, light, heat, oxygen, even environmental stresses such as heavy metals, bacteria, and fungi on low-cost culture media must be investigated and established in order to optimize the production.

ACKNOWLEDGMENTS

This study was financially supported by the Natural Science Foundation of Zhejiang Province (No. LR13C200002) and the National Nature Science Foundation of China (No. 31871904). The authors thank Dr. Wei Zhang of the Lerner Research Institute, Cleveland, OH, USA, for revising the manuscript and polishing the language.

AUTHOR CONTRIBUTIONS

Hongyun Lu collected and analyzed the data and drafted the manuscript. Hanghang Lou and Zhengjie Liu participated in the final manuscript review and editing. Jingjin Hu wrote the complementary paragraphs. Qihe Chen critically analyzed the data and revised the entire manuscript. All authors revised the manuscript and approved the final version.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

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How to cite this article: Lu H, Lou H, Hu J, Liu Z, Chen Q. Macrofungi: A review of cultivation strategies, bioactivity, and application of mushrooms. *Compr Rev Food Sci Food Saf*. 2020;1–24. https://doi.org/10.1111/1541-4337.12602